GENERAL ORGANIC CHEMISTRY — PART 1

STRUCTURE · BONDING · ELECTRONIC EFFECTS · AROMATICITY

1 HYBRIDISATION

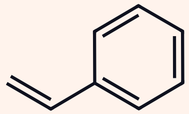
sp^3 / sp^2 / sp · geometry · steric-number rule · allene & cumulenes

2 σ & π BONDS

overlap type · bond length · bond enthalpy · % s-character

3 BOND-LINE & DoU

skeletal drawing · σ/π counting · degree of unsaturation



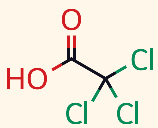
4 BOND FISSION

homolytic vs heterolytic · fish-hook & full curved arrows

A · B → A• + •B

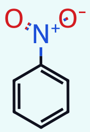
5 INDUCTIVE EFFECT

−I / +I orders · distance decay · acidity & basicity applications



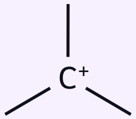
6 RESONANCE (+M / −M)

rules of contributors · resonance energy · steric inhibition



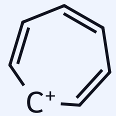
7 HYPERCONJUGATION

α-H count · no-bond resonance · alkene & cation stability



8 AROMATICITY

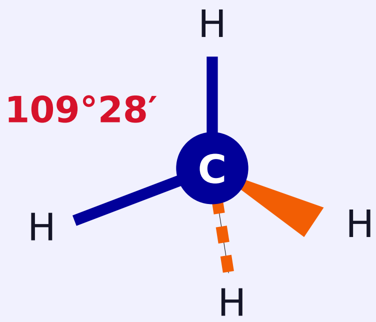
Hückel 4n+2 · anti- vs non-aromatic · Frost circle · ions



★ = **MUST-MEMORISE ORDER** (−I, +I, −M, +M, stability, acidity, basicity) — learn every starred box by heart. **Red boxes = JEE traps.**

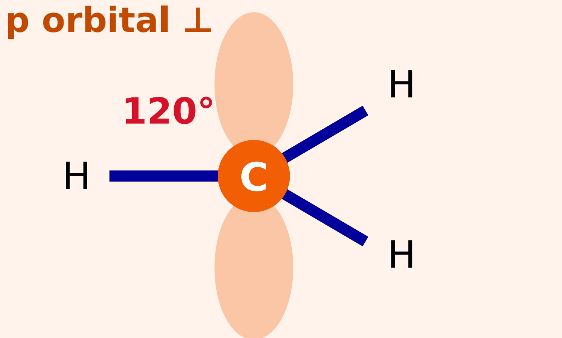
1 THE THREE HYBRID STATES OF CARBON

sp^3 · TETRAHEDRAL



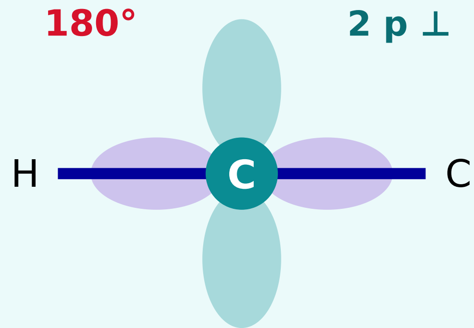
25 % s · 4 σ · 0 π

sp^2 · TRIGONAL PLANAR



33.3 % s · 3 σ · 1 π

sp · LINEAR



50 % s · 2 σ · 2 π

60-SECOND RULE — STERIC NUMBER

SN = (σ bonds) + (lone pairs)

SN	Hybrid	Shape	Angle
4	**sp^3**	tetrahedral	109°28′
3	**sp^2**	trigonal planar	120°
2	**sp**	linear	180°

π bonds **never** counted

SPOT THE HYBRIDISATION

CH_4

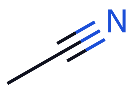
CH_4 — **sp^3**

C_2H_4 — **sp^2**

C_2H_2 — **sp**

all C **sp^2**

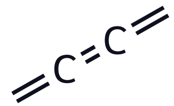
C=O carbon **sp^2**



C≡N carbon **sp**

CUMULENES — THE CLASSIC TRAP

allene · 2 C=C



butatriene · 3 C=C

C=C count EVEN → ends ⊥ | ODD → ends planar

- Central C **sp** · ends **sp^2**
- ⊥ ends ⇒ **axial chirality**

BOND ANGLE — QUICK COMPARISONS

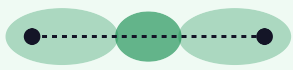
- ★ **CH_4 109.5° > NH_3 107° > H_2O 104.5°** **(lp repulsion)**
- ★ **Ring: cyclopropane 60° (banana) < cyclobutane 88° < cyclopentane 108°**

TRAPS

- ✗ Lone pair conjugated to π ⇒ that atom is **sp^2** (amide N, phenol O, aniline N)
- ✗ Carbocation C is **sp^2**; carbanion C is **sp^3** pyramidal

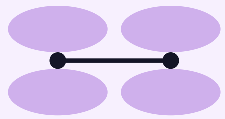
2 OVERLAP GEOMETRY DECIDES EVERYTHING

σ — HEAD-ON (axial)



strong · free rotation · can exist alone

π — SIDE-ON (lateral)



weak · no rotation · only with a σ

★ **% s** **sp (50 %) > sp² (33.3 %) > sp³ (25 %)**

as % s ↑ · bond **shorter** & **stronger** · carbon **more electronegative** (sp 3.29 > sp² 2.75 > sp³ 2.48) · attached H **more acidic** · lone pair **less basic** · bond angle **larger**

BOND PARAMETERS — LEARN THIS TABLE

Bond	Length (Å)	Mean bond enthalpy (kJ mol⁻¹)	Order
C–C	1.54	348	1
C=C	1.34	614	2
C≡C	1.20	839	3
C–C benzene	**1.39**	—	**1.5**
C–H (sp³)	1.09	414	1
C–H (sp²)	1.086	—	1
C–H (sp)	**1.06**	—	1

π weaker than σ: E(π) ≈ 614 − 348 = **266** < 348 ⇒ π reacts first

% s-CHARACTER — THE MASTER CHAIN

↑ % s ⇒ e⁻ pair held closer:

Property	Trend with ↑ % s
Bond length	**decreases**
Bond strength	**increases**
Electronegativity of C	**increases**
Acidity of attached H	**increases**
Basicity of lone pair	**decreases**
Bond angle	**increases**

IMMEDIATE PAY-OFF

Acidity of terminal H

HC≡CH	H₂C=CH₂	H₃C–CH₃
pK_a **25**	pK_a **44**	pK_a **50**

★ **sp > sp² > sp³**

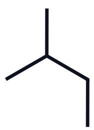
Basicity of N lone pair

★ **R_3N > pyridine > R–C≡N**

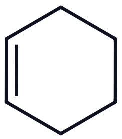
Note: "more acidic than ethene" ≠ acidic in water

3 BOND-LINE (SKELETAL) FORMULA — 4 RULES

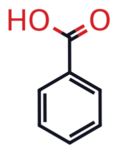
- Every **vertex** / **line end** = carbon
- C–H never drawn — fill valency to 4
- **Heteroatoms** + their H always written
- Charges & odd e⁻ shown explicitly



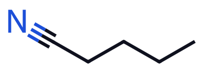
C_5H_{12}



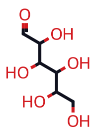
C_6H_{10}



$C_7H_6O_2$



pentanenitrile



glucose (open chain)

COUNTING σ AND π — ONE UNIVERSAL FORMULA

σ = (total atoms − 1) + (number of rings)

π = (number of C=C) + 2 × (number of C≡C) + …

Family	Formula	σ	π
Alkane	C_nH_{2n+2}	**3n + 1**	0
Cycloalkane	C_nH_{2n}	**3n**	0
Alkene (open)	C_nH_{2n}	**3n − 1**	1
Alkyne (open)	C_nH_{2n-2}	**3n − 3**	2
Benzene	C_6H_6	**12**	**3**

Benzene: 12 atoms, 1 ring ⟹ σ = 11 + 1 = **12** ✓

DEGREE OF UNSATURATION (DoU)

$$\text{DoU} = \frac{2C + 2 + N - H - X}{2}$$

DoU = rings + π bonds

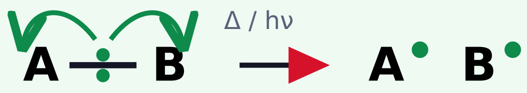
O, S **ignored** · X = halogen · each N adds +1

Compound	DoU	= rings + π
C_6H_6 benzene	**4**	1 + 3
$C_{10}H_8$ naphthalene	**7**	2 + 5
C_6H_5Cl	**4**	1 + 3
C_6H_7N aniline	**4**	1 + 3
$C_6H_{12}O_6$ glucose	**1**	0 + 1 (C=O)

Trap: DoU ≥ 4 *hints* at benzene, never proves it

4 HOW A COVALENT BOND BREAKS

HOMOLYTIC (symmetrical)



FREE RADICALS

half-headed "fish-hook" arrow ↷ 1 e^-

non-polar bond · non-polar solvent (CCl_4)

peroxide, UV, high T, gas phase

HETEROLYTIC (unsymmetrical)

CATION + ANION

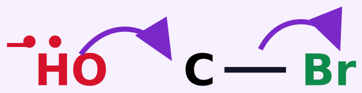
full-headed curved arrow ↷ 2 e^-

polar bond · polar solvent (H_2O, ROH)

e^- pair → more electronegative atom

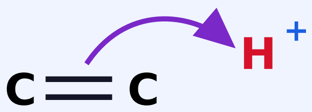
CURVED-ARROW NOTATION (ELECTRON PUSHING)

LONE PAIR → σ*



tail = e^- rich · head = e^- poor

π BOND → E⁺



π e^- attack the electrophile

3 GOLDEN RULES

① Arrows push **electrons**, never atoms

② Never exceed the **octet** for C,N,O,F

③ Net **charge is conserved**

start: lone pair / π / σ / − → end: + / atom

ARROW TAIL ALWAYS STARTS AT AN e^- SOURCE

Source (tail)	Sink (head)
lone pair on O, N, S, X^-	C of C=O, C^+, H of H–X
π bond (C=C, C=O)	E^+, H^+, Br^+, NO_2^+
σ bond (C–H, C–C in H.C.)	adjacent empty p / π*
negative charge	positive charge

WHICH FISSION WILL HAPPEN?

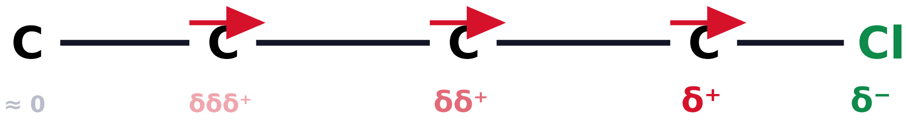
	Homolytic	Heterolytic
Bond	non-polar	polar
Solvent	CCl_4, hexane	H_2O, ROH, DMSO
Trigger	hν, Δ, peroxide	Lewis acid / base
Arrow	half (fish-hook)	full (double)
Product	radicals	ions
Energy	BDE	higher

THE THREE FATAL MISTAKES

1. Arrow must start at the **base's lone pair** — never at H
2. Carbon with **5 bonds**. Octet is capped: a bond forms ⇒ one breaks
3. ↔ is resonance, not reaction. ↔ **never moves atoms**

5 PERMANENT σ-BOND POLARISATION · TRANSMITTED THROUGH THE CHAIN · DIES OUT WITH DISTANCE

Falls ≈ ⅓ per carbon → dead after 3 atoms



C-4 | C-3 (≈1/27) | C-2 (≈1/9) | C-1 (≈1/3)

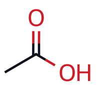
★ **−I** **−N⁺R₃ > −NO₂ > −CN > −SO₃H > −CHO > −COOH > −F > −Cl > −Br > −I > −OR > −OH > −NH₂ > −C₆H₅ > −H**

ELECTRON WITHDRAWING · acidity ↑ basicity ↓ · halogens follow **electronegativity** (the *opposite* of their +M order) · also −I, in place : $-SO_2R$ (after $-NO_2$) · −CO−, −COCl, $-CONH_2$ (near −COOH) · −C≡CH, $-CH{=}CH_2$ (weak)

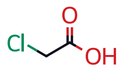
★ **+I** **$-O^-$ > $-COO^-$ > $-C(CH_3)_3$ > $-CH(CH_3)_2$ > $-CH_2CH_3$ > $-CH_3$ > −H**

ELECTRON DONATING · acidity ↓ basicity ↑ · only a **full ⊖ charge** and **alkyl** groups are +I · **stabilises** carbocations, **destabilises** carbanions

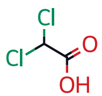
ACIDITY — THE SHOWCASE (pK a)



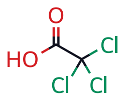
CH_3COOH
4.76



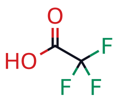
$ClCH_2COOH$
2.86



$Cl_2CHCOOH$
1.29



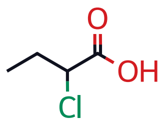
Cl_3CCOOH
0.65



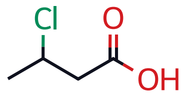
F_3CCOOH
0.23

more −I groups → weaker conjugate base → lower pK_a

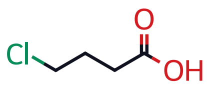
DISTANCE MATTERS — THE 1/3 RULE



α (2-Cl)
pK_a **2.86**



β (3-Cl)
pK_a **4.05**



γ (4-Cl)
pK_a **4.52**

butanoic acid 4.82 · **dead beyond C-3**

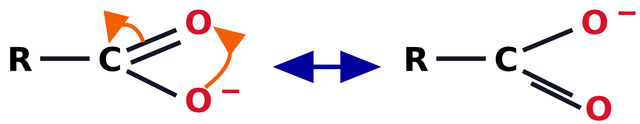
SIGNATURE OF THE INDUCTIVE EFFECT

- **Permanent**, weak, no reagent
- Through **σ bonds only**
- Only **partial** charges δ^+/δ^-
- **Additive** — 3 Cl beat 1 Cl
- ≈ ⅓ per carbon ⇒ dead after 3

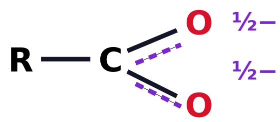
MUST-KNOW APPLICATIONS

- Carbocation stability: 3° > 2° > 1° > CH_3^+
- Carbanion stability: **reverse** — CH_3^- > 1° > 2° > 3°
- HCOOH 3.75 > CH_3COOH 4.76 (no +I alkyl)
- Oxalic pK_{a1} 1.23 — 2nd −COOH is −I
- Gas phase amines: 3° > 2° > 1° > NH_3

6 DELOCALISATION OF π / LONE-PAIR ELECTRONS · ↔ IS NOT EQUILIBRIUM — THE REAL MOLECULE IS ONE HYBRID



RESONANCE HYBRID



both C–O bonds identical = 1.27 Å

★ **+M / +R** −O⁻ > −NH₂ ≈ −NR₂ > −OH ≈ −OR > −NHCOR > −OCOR > −C₆H₅ > −F > −Cl > −Br > −I

push e⁻ **INTO** the ring · **o,p**-directing · activating · halogen +M follows **2p-2p overlap**, so −X stays **deactivating** (−I > +M) yet still directs **o,p**

★ **−M / −R** −NO₂ > −CN > −SO₃H > −CHO > −CO− > −COOH > −COOR > −CONH₂ > −COO⁻

pull e⁻ **OUT** of the ring · **m**-directing · deactivating · the group must have a π bond to an electronegative atom

4 CONDITIONS FOR RESONANCE

- ✔ **Conjugation** on **alternate** atoms
- ✔ **Planar** — parallel p orbitals
- ✔ **Nuclei fixed** — only e⁻ move
- ✔ **Same unpaired** e⁻ count

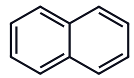
Trap: sp³ CH₂ between two π systems **kills** conjugation

RANKING THE CONTRIBUTORS (stability rules)

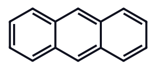
- ▸ ① More **covalent bonds** wins
- ▸ ② **Complete octet** wins
- ▸ ③ **No charge separation** wins
- ▸ ④ ⊖ on **more** EN · ⊕ on **less** EN
- ▸ ⑤ Like charges **apart**, unlike **close**
- ▸ ⑥ **Aromatic** contributor dominates

RESONANCE ENERGY (kJ mol⁻¹)

benzene
150 (36 kcal)



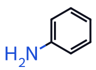
naphthalene
255



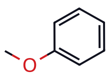
anthracene
351

RE / ring falls 150 > 127 > 117 ⇒ reactivity **rises**

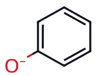
+M / +R (push e⁻ INTO the ring · o,p-directing · activating)



aniline



anisole

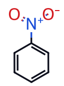


phenoxide

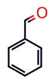
vinyl chloride

F > Cl > Br > I (2p–2p) · net **deactivating** but **o,p**-directing

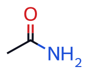
−M / −R (pull e⁻ OUT of the ring · m-directing · deactivating)



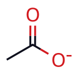
nitrobenzene



benzaldehyde



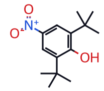
amide (N is sp²)



carboxylate

Amide: N lp → O ⇒ N **sp²**, C–N partly double, **very poor base**

STERIC INHIBITION OF RESONANCE (SIR)

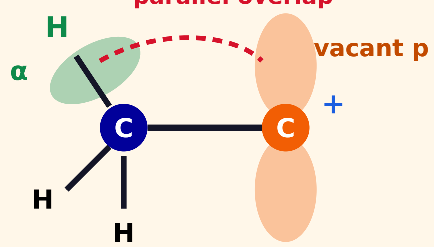


2,6-di-*t*-butyl-4-nitrophenol

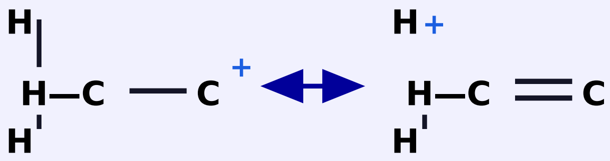
Bulky *ortho* groups twist −NO₂ / −NR₂ **out of plane** ⇒ resonance switched **OFF**

7 σ(C-H) OF AN sp^3 CARBON DELOCALISES INTO AN ADJACENT EMPTY p / π / ODD ELECTRON

σ(C-H) → EMPTY p (carbocation)



"NO-BOND RESONANCE"



H is **not** actually free — the σ pair is shared

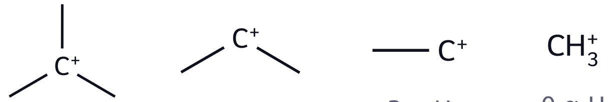
THE ONLY RULE YOU NEED

Stability ∝ number of α-hydrogens

α-H = H on the sp^3 C **next to** the ⊕ centre

- ✔ Stronger than −I, weaker than resonance
- ✘ Impossible if there is **no α-H** (vinyl⁺, phenyl⁺, CH_3^+)
- ✘ Impossible if the α-C is sp^2 or sp

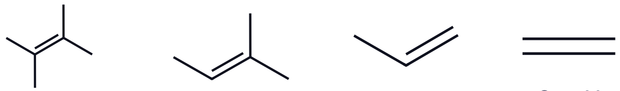
CARBOCATION STABILITY BY α-H COUNT



9 α-H **3°** | 6 α-H **2°** | 3 α-H **1°** | 0 α-H **CH_3^+**

★ **3° > 2° > 1° > CH_3^+**

ALKENE STABILITY (Zaitsev's real reason)



12 α-H tetra-subst. | 9 α-H tri-subst. | 3 α-H mono | 0 α-H ethene

★ **tetra > tri > di > mono > ethene**

STRUCTURAL EVIDENCE — PROPENE

Bond	Normal	In propene
C1=C2	1.34 Å	**1.36 Å ↑**
C2–C3	1.54 Å	**1.488 Å ↓**

Single bond **shortens**, double **lengthens** ✓

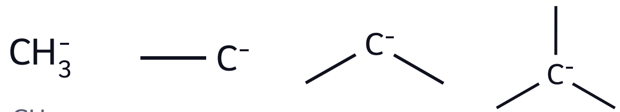
OTHER APPLICATIONS

- **Free radicals**: 3° > 2° > 1° > CH_3•
- **$-CH_3$ is o,p-directing** and activating (hyperconjugation + weak +I)
- Propene μ = 0.35 D — hyperconjugation
- **Reverse hyperconjugation** in $-CF_3$ / $-CCl_3$ ⇒ order **inverts**

THE BIG EXCEPTION — CARBANIONS

Hyperconjugation DESTABILISES carbanions — lp / σ(C–H) repulsion:

★ **CH_3^- > 1° > 2° > 3° carbanion**



CH_3^- 0 α-H | 1° 3 α-H | 2° 6 α-H | 3° 9 α-H

8 ELECTROMERIC EFFECT (E) — TEMPORARY & REAGENT-DRIVEN

Complete transfer of a **π pair** the moment a reagent approaches · **reversible**

+E (toward the attacking atom)

$C{=}C + H^+ \longrightarrow C^+{-}C{-}H$

electrophilic addition

−E (away from the attacking atom)

$C{=}O + CN^- \longrightarrow NC{-}C{-}O^-$

nucleophilic addition

THE FOUR EFFECTS SIDE BY SIDE

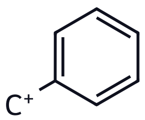
Feature	Inductive (I)	Resonance (M)	Hyperconj. (H)	Electromeric (E)
Bond involved	σ	π / lone pair	σ(C–H) → p/π	π
Nature	permanent	permanent	permanent	**temporary**
Reagent needed	no	no	no	**YES**
Charge produced	partial δ	full ± on hybrid	partial	**full ±**
Distance	dies in 3 atoms	whole conjugated system	α position only	at the double bond
Planarity	not required	**required**	required	required

★ **DOMINANCE** **RESONANCE (M) > HYPERCONJUGATION > INDUCTIVE (I)**

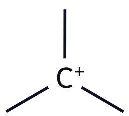
electromeric (E) acts **only** while a reagent is present — and then it overrides all · **aromaticity beats every one of them**

ORDER OF DOMINANCE — DECIDING STABILITY

E acts **only** with a reagent — then it overrides all



benzyl⁺ (resonance) >



3° (hyperconj.) >

1° (inductive)

WHEN INDUCTIVE BEATS RESONANCE — THE HALOGEN PARADOX

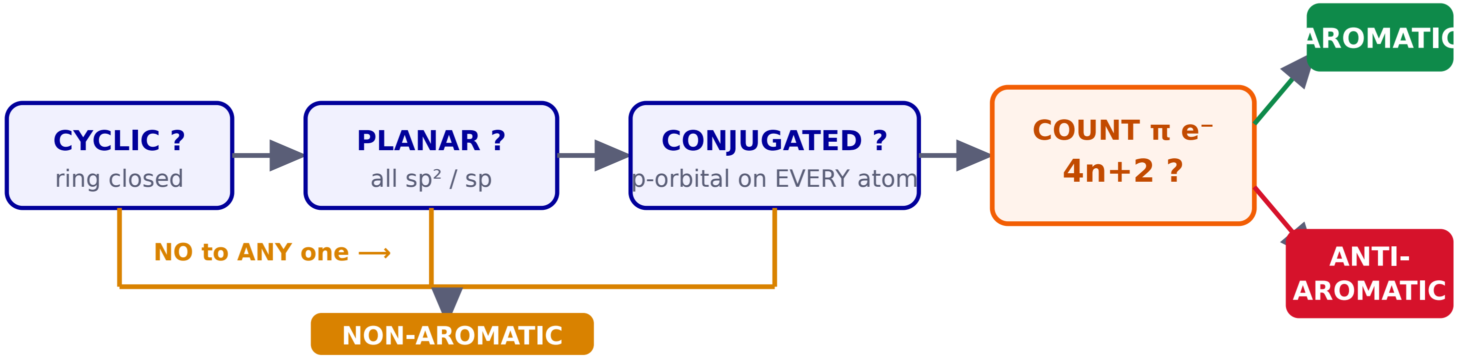
Question	Which effect wins	Answer
Ring **reactivity** (rate)	**−I** (stronger)	deactivating
Ring **orientation**	**+M**	**o, p-** directing

Most-asked trap: **−X deactivates BUT directs o,p**

RAPID-FIRE VERDICTS

- Phenol ≫ ethanol → **resonance**
- CH_3COOH > EtOH → **resonance**
- Aniline ≪ NH_3 → **resonance**
- Alkene stability → **hyperconjugation**
- Cl_3CCOOH > CH_3COOH → **−I**
- Tropylium → **aromaticity**

9 THE FOUR-STEP TEST — APPLY IN ORDER



HÜCKEL'S RULE

π electrons = 4n + 2   n = 0, 1, 2, 3 …

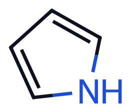
2, 6, 10, 14, 18 …

Class	π e⁻	Stability
AROMATIC	4n + 2	extra stable
ANTIAROMATIC	4n	destabilised
NON-AROMATIC	fails a test	about the same

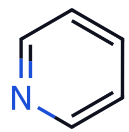
Escape: antiaromatic rings **pucker** to become non-aromatic (COT is tub-shaped)

COUNTING π ELECTRONS — 4 RULES

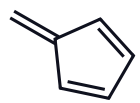
- ① Each ring C=C contributes **2**
- ② A ⊕ charge (empty p) contributes **0**; a ⊖ charge (lone pair in p) contributes **2**
- ③ Heteroatom lp counts **only if** that atom has no ring π bond
- ④ **Exocyclic** C=C contributes **0**



pyrrole **6π**
lp counted



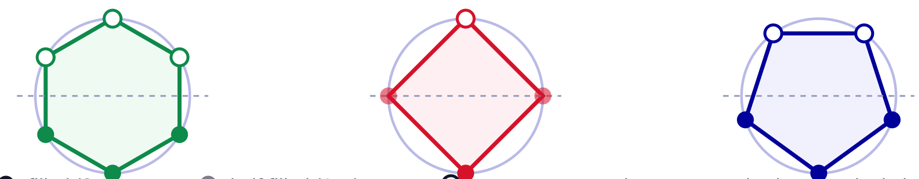
pyridine **6π**
lp NOT counted



fulvene
non-aromatic

MOLECULAR-ORBITAL PROOF — FROST CIRCLE

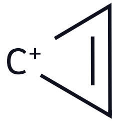
FROST CIRCLE — inscribe polygon vertex-DOWN; each vertex = an MO level



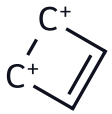
● filled (2e⁻)   ● half-filled (1e⁻)   ○ empty   · lowest vertex is always a single bonding MO

benzene 6 π   **cyclobutadiene 4 π**   **$C_5H_5^-$ 6 π**

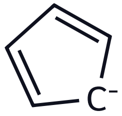
AROMATIC (4n + 2)



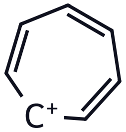
cyclopropenyl ⊕
2π · n=0



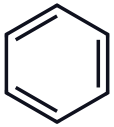
cyclobutadienyl **2**⊕
2π

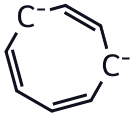


cyclopentadienyl ⊖
6π

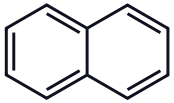


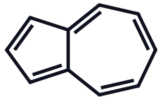
tropylium ⊕
6π

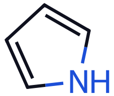

benzene
6π



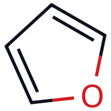
COT **2**⊖
10π · planar


naphthalene
10π


azulene
10π · μ = 1.08 D

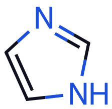


pyrrole 6π

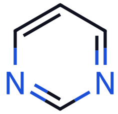


furan 6π

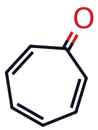
thiophene 6π



imidazole 6π

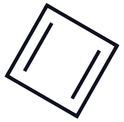


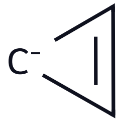
pyrimidine 6π



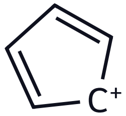
tropone 6π

ANTIAROMATIC (4n)

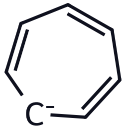

cyclobutadiene
4π



cyclopropenyl ⊖
4π



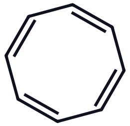
cyclopentadienyl ⊕
4π

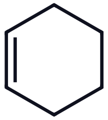


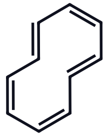
tropylium ⊖
8π

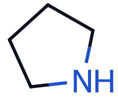
Same ring, opposite verdict: $C_5H_5^+$ least stable cation · $C_5H_5^-$ most stable anion

NON-AROMATIC


COT
tub-shaped, 8π


cyclohexene
sp³ present


[10]annulene
10π but **non-planar**



pyrrolidine
no conjugation

[10]annulene: 4n+2 ✓ but **non-planar** (H–H clash) ⇒ check planarity **first**

ANNULENES — THE PLANARITY FILTER

Annulene	π e⁻	Planar?	Verdict
[4]	4	yes	antiaromatic
[6] benzene	6	yes	aromatic
[8] COT	8	**no** (tub)	non-aromatic
[10]	10	**no** (H clash)	non-aromatic
[14]	14	yes	aromatic
[18]	**18**	yes — big enough	aromatic

EXAM CONSEQUENCES OF AROMATICITY

- Undergoes **substitution**, resists addition
- All ring C–C equal (benzene 1.39 Å)
- **Ring current** ⇒ downfield ¹H NMR (δ 7–8)
- **cyclopentadiene pK_a = 16** ≈ water
- Aromatic contributor **dominates** the hybrid
- Hückel is for **monocyclic** rings